

MSCF Probability: Homework Set #1 Solutions

1. 6 points (4 for set-up, 2 for work and answer)

Let C, F_1, F_2 , and F_3 represent the following events:

C : Member has hospital expenses exceeding \$100,000.

F_1 : Member is under the age of 40.

F_2 : Member is between the age of 40 and 65.

F_3 : Member is over 65.

From the given information, we have

$$P(C|F_1) = 0.03, \quad P(C|F_2) = 0.04, \quad P(C|F_3) = 0.10, \quad P(F_3) = 0.20.$$

To avoid paying an additional reinsurance premium, the HMO requires that

$$P(C) \leq 0.05.$$

By the law of total probability,

$$\begin{aligned} P(C) &= P(C|F_1)P(F_1) + P(C|F_2)P(F_2) + P(C|F_3)P(F_3) \\ &= (0.03)P(F_1) + (0.04)P(F_2) + (0.10)(0.20) \\ &= (0.03)P(F_1) + (0.04)P(F_2) + 0.02. \end{aligned}$$

Hence, the requirement $P(C) \leq 0.05$ is equivalent to

$$(0.03)P(F_1) + (0.04)P(F_2) \leq 0.03;$$

that is,

$$3P(F_1) + 4P(F_2) \leq 3.$$

Put $x = P(F_1)$. Then $P(F_2) = 1 - P(F_1) - P(F_3) = 1 - x - 0.2 = 0.8 - x$. Thus, we have

$$3x + 4(0.8 - x) \leq 3,$$

and

$$x \geq 0.2$$

Hence, to avoid exceeding the 5% trigger point, at least 20% of the HMO's members should be under the age of 40.

2. 4 points

Let C_1, C_2 , and C_3 denote the events of drawing the loaded heads coin, loaded tails coin, and fair coin, respectively. Let H_i be the event of getting heads on the i th toss. So, we want

$$\begin{aligned} \Pr(C_3|H_1) &= \frac{\Pr(H_1|C_3) \Pr(C_3)}{\Pr(H_1|C_1) \Pr(C_1) + \Pr(H_1|C_2) \Pr(C_2) + \Pr(H_1|C_3) \Pr(C_3)} \\ &= \frac{\left(\frac{1}{2}\right) \left(\frac{1}{3}\right)}{\left(1\right) \left(\frac{1}{3}\right) + \left(0\right) \left(\frac{1}{3}\right) + \left(\frac{1}{2}\right) \left(\frac{1}{3}\right)} \\ &= \frac{1}{6} \cdot \frac{6}{3} = \frac{1}{3}. \end{aligned}$$